भारतीय प्रौद्योगिकी संस्थान कानपुर
Indian Institute of Technology Kanpur

MATHEMATICS

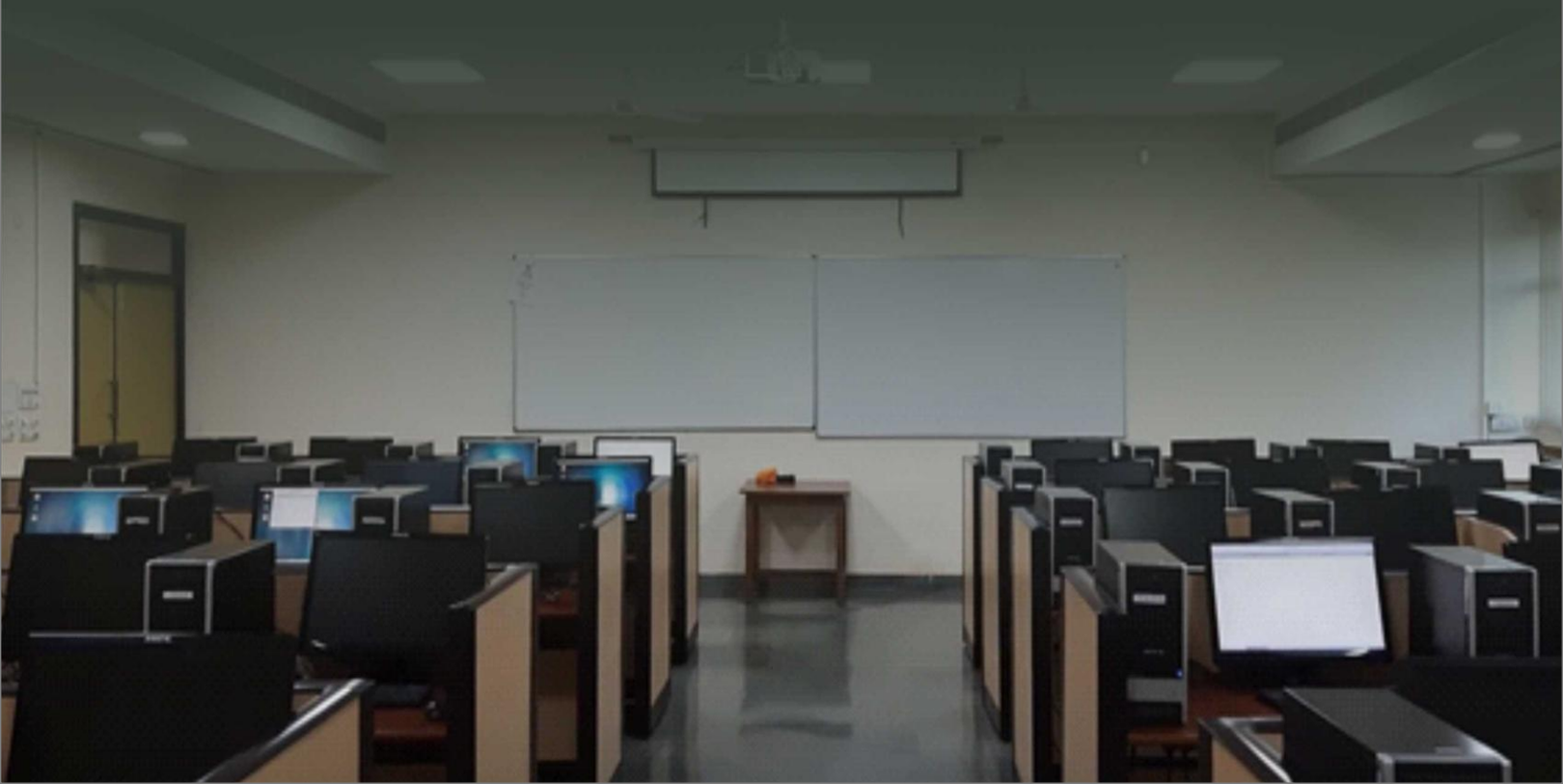
POST GRADUATE PROGRAM

Website: www.iitk.ac.in/math



MATHEMATICS

The Department offers PhD programs in Mathematics with the objective of preparing motivated researchers who can work in frontline areas as well as to prepare well-trained teaching faculty. The vibrant academic environment of the department is nurtured by strongly motivated faculty. The department faculty takes research initiatives to work in recent and emerging areas, and encourages interdisciplinary trends. Around 440 research scholars have graduated from the department and are serving in the faculty of various institutes and universities in India and abroad. Some of them are in industry as well.



PHD PROGRAMME OFFERED

- PhD in Mathematics



Lab Facilities

In addition to the Computer Centre (CC) of the institute which caters to specialized computational needs of the academic community, the department has its own computing facilities for faculty, staff and students of the department as follows:

- A Linux lab having 48 high end desktops with remote access facility;
- A Windows lab having 34 high end desktops with remote access facility;
- Advanced GPU and non-GPU workstations;
- HPC clusters installed in centralized data centers: 2 master nodes, 20 nodes;
- Separate UPS for network racks and switches;
- Lab desktops connected with centralized UPS having a good backup

Our students can access various licensed software available from the CC. They are also provided with accessibility to many ebooks, reputable journals, Overleaf, etc., through ONOS and Institute subscriptions. Additionally, access to high-speed internet.

(list of applications: https://www.iitk.ac.in/cc/softwares)

Computer Labs

- One Linux lab is well equipped with 48 desktop computers;
- One Windows lab is well equipped with 34 desktop computers;
- 3 rooms for postdoc fellows with personal cabin;
- 5 research scholar rooms, each scholar gets a personal computer and a cubicle;
- All PCs have advanced configurations (Core i7/Core i5) with latest software;
- Labs are maintained by CC engineers and supporting staff.

Infrastructure for Seminars

- Three seminar rooms equipped with projectors, screen and multimedia;
- For interviews/seminars, we have Zoom/Skype video conferencing facilities

Printing Facility

- Our department has 3 high-end photocopier machines with scanning and printing facilities for department academic courses.
- We also provide printing and scanning facility to our PhD students for their research. For UG and PG students, this is on a request basis.



FACULTY LIST

- ANAND A. (PhD, University of Minnesota), Research Interests: Computational methods for the solution of partial differential and integral equations, with emphasis on high-order efficient simulators.
- BAHUGUNA, D. (PhD, IIT Kanpur), Research Interests: Differential Equations, Non-linear Analysis, Theory of Semigroups.
- BAKSHI, KESHAB CHANDRA (PhD, IMSc, Chennai), Research Interests: C*-algebras and von Neumann algebras. In particular, Subfactor theory and Planar algebras.
- BAL, KAUSHIK (PhD, UPPA, France), Research Interests: Elliptic and Parabolic Partial Differential Equations.
- BANERJEE, MALAY (PhD, University of Calcutta), Research Interests: Mathematical Ecology and Eco-epidemiology, Nonlinear Dynamics.
- BANERJEE, MOHUA (PhD, University of Calcutta), Research Interests: Rough Set Theory and its Applications, Modal Logics.
- Bera, Sayani (PhD, IISc Bangalore), Research Interests: Holomorphic dynamics, Several Complex Variables
- BISWAS, ABHIJIT (PhD, Temple University), Research Interests: Numerical methods for differential equations, time integration, hyperbolization.
- BISWAS, MRINMAY (PhD, IISER Kolkata), Research Interests: PDE, Stochastic PDE, Control Theory and Numerics.
- BONALA, NARASIMHA CHARY (PhD, CMI, Chennai), Research Interests: Algebraic Groups, Algebraic geometry and Representation Theory
- CHANDEL, VIKRAMJEET SINGH (PhD, IISc, Bengaluru), Research Interests: Several Complex Variables.
- CHAVAN, SAMEER (PhD, University of Pune), Research Interests: Operator Theory.
- CHOWDHURY, INDRANIL (PhD, TIFR - CAM, Bengaluru), Research Interests: Theory and Numerical Analysis of PDEs, Nonlocal/ Fractional order Problems, Fully Nonlinear Equations.
- DAR, APARNA (PhD, S.U.N.Y Stony Brook, USA), Research Interests: Differential Geometry, Algebraic Topology, Knot Theory.
- GANGULY, ARIJIT (PhD, TIFR Mumbai), Research Interests: Diophantine approximation, Dynamical systems and Random walks.

FACULTY LIST

- GHORAI, S. (PhD, Leeds, UK), Research Interests: Computational Fluid Dynamics, Mathematical Biology, Adaptive Unstructured grid.
- GUIN, SATYAJIT (PhD, IMSc, Chennai), Research Interests: Noncommutative Geometry, Operator Algebra.
- JHA, SOMNATH (PhD, Tata Institute of Fundamental Research, Mumbai), Research Interests: Number Theory, Arithmetic Geometry.
- KUBER, AMIT (PhD, University of Manchester, UK), Research Interests: Representation theory of associative algebras, linear orders and combinatorics of words, model theory.
- Kumar, Ashutosh (PhD: University of Wisconsin Madison), Research Interests: Mathematical Logic, Set Theory and its applications to Analysis, Measure Theory and Topology.
- MALOO, ALOK KUMAR (PhD, TIFR, Mumbai), Research Interests: Commutative Algebra.
- MANDAL, ASHIS (PhD, ISI, Kolkata), Research Interests: Algebraic Topology; Deformation of Algebraic Structures, Higher Structures and Related Fields.
- MOHANTY, PARASAR (PhD, IIT Kanpur), Research Interests: Harmonic Analysis
- MUTHUKUMAR, P. (PhD, Indian Statistical Institute, Chennai), Research Interests: Complex Analysis and Operator Theory
- MUTHUKUMAR, T. (PhD, Institute of Mathematical Sciences), Research Interests: Homogenization and Variational Methods for PDE's, Elliptic PDE's, Optimal controls.
- NADIMPALLI, SANTOSH, Research Interests: Representation theory.
- NILAKANTAN, NANDINI (PhD, IISc, Banglore), Research Interests: Combinatorial Topology, Computational Geometry.
- PAL, ABHIJIT (PhD, Indian, Statistical Institute, Kolkata), Research Interests: Geometric Group Theory, Relatively Hyperbolic Groups, Mapping Class Groups.
- PATNAIK, S. (PhD, University of Cincinnati, USA), Research Interests: Operator Theory
- PATTANAYAK, S. K. (PhD, CMI, Chennai), Research Interests: Algebraic Groups and Invariant Theory, Lie Algebras and Representation Theory.
- RATHISH KUMAR, B.V. (PhD, SSSIHL, prasanthi nilayam), Research Interests: Parallel Computing, Numerical Methods For PDEs, Computational Fluid Dynamics, Finite Element Analysis, Computational Biomechanics, Porous Media, Image Processing

FACULTY LIST

- RAWAT, RAMA (PhD, ISI, Bangalore), Research Interests: Harmonic analysis on Lie groups.
- ROY, PROSENJIT (PhD, University of Zurich, Switzerland), Research Interests: Partial Differential Equations.
- SAMUEL, PREENA (PhD, IMSc, Chennai), Research Interests: Structure theory and representations of algebraic groups.
- SANKI, BIDYUT (PhD, Indian Institute of Science, Bangalore), Research Interests: Topology and Geometry.
- SEN DEBASIS (PhD, ISI, Kolkata)), Research Interests: Homotopy theory, Group actions.
- SHARMA, SACHIN (PhD, IMSc, Chennai), Research Interests: Representation theory of Kac-Moody algebra, extended affine Lie algebra.
- SHUNMUGARAJ, P. (PhD, IIT Bombay), Research Interests: Functional Analysis.
- SHEKHAR, SUDHANSHU (PhD, TIFR, Mumbai), Research Interests: Algebraic Number Theory and Arithmetic Geometry.
- SINGH, SAURABH KUMAR (PhD, TIFR, Mumbai), Research Interests: L-Function, Sub-convexity problems, Sieve Method, Riemann Zeta Function.
- Singla, Pooja (PhD, IMSc, Chennai), Research Interests: Representations theory of finite and p-adic groups, Applications of representation theory.
- THAKUR, AJAY SINGH (PhD, IMSc, Chennai), Research Interests: Algebraic Topology, Complex Manifolds.
- Vedansh, Arya (PhD, TIFR-CAM), Research Interests: Analysis of PDEs, Regularity theory for elliptic and parabolic PDEs, Unique continuation problems, Free boundary problems, and Curvature driven flows.

BROAD RESEARCH AREAS

- Algebraic geometry
- Commutative Algebra
- Complex Analysis & Operator Theory
- Computational Acoustics and Electromagnetics
- Computational Fluid Dynamics
- Data Mining in Finance
- Differential Equations
- Entropy Estimation and Applications
- Ergodic Theory and Dynamical Systems
- Functional Analysis & Operator Theory
- Game Theory
- Harmonic Analysis
- Homological Algebra
- Image Processing
- Mathematical Biology
- Number Theory & Arithmetic Geometry
- Numerical Analysis and Scientific Computing
- Operator Algebra
- Regularity theory of Partial Differential Equations
- Representation Theory
- Set Theory and Logic
- Several Complex Variables
- Step-Stress Modelling
- Stochastic Partial Differential Equations
- Topology and Geometry
- Tribology

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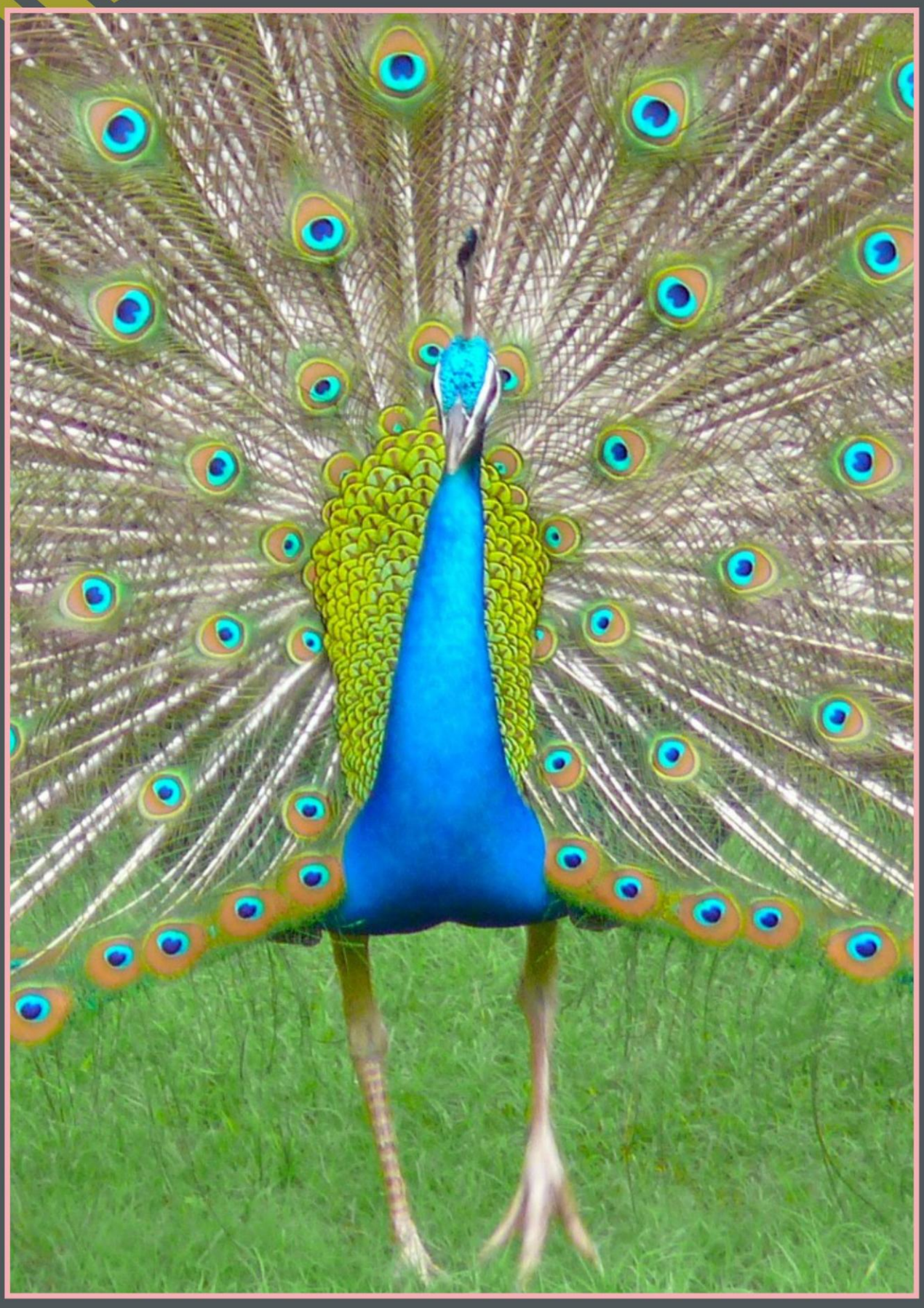
CONTACT

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